PROJECT NAME : Retail Sales Analysis using Mysql On Superstore

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REAL CANADIAN
SUPERSTORE

INTRODUCTION

This project focuses on analyzing historical retail sales data from a fictional Superstore using MySQL. The dataset contains information about orders, products, customers, shipping methods, and geographic locations. By using SQL queries, we extract meaningful insights to support business decisions such as improving sales, identifying high-performing regions and customers, and understanding profit trends.

PROJECT DESCRIPTION

The **Superstore** dataset simulates a real-world business scenario in the retail
sector. The goal of the project is to perform **data exploration and analysis**
using SQL to uncover trends and patterns in:

Sales and profit over time

Customer segmentation and top buyers

Product category performance

Regional growth and profitability

Shipping mode efficiency

Discount impacts on profit

PROJECT PROBLEM STATEMENT :

Retail businesses collect large amounts of data from their daily operations, but they often struggle to turn this data into actionable insights. The main challenges addressed in this project include:

Key Problems:

1. **Lack of visibility into product performance:**

   o Which categories and sub-categories generate the most sales or profit?

2. Inability to track customer behavior:

   o Who are the most valuable customers?

   o Which customer segments contribute the most to profit?

3. No clear understanding of geographic performance:

- o Which cities or regions are the most profitable?

4. Limited insight into discount strategy:

- o Are discounts increasing sales but reducing profit?

5. Inefficient shipping methods:

- o Which shipping methods are used the most?
- o Do they affect delivery times and profits?

Solution Using MySQL:

Using SQL queries, we address the above challenges by:

Aggregating and filtering the data

Grouping by key business dimensions (product, customer, region, etc.)

Calculating KPIs (Sales, Profit, Quantity)

Visualizing trends (optional using BI tools later)

QUESTIONS

Q.1 Which cities or regions are the most profitable ?

Q.2 What are the total profits and total sales per city ?

Q.3 What region generates the highest sales and profits ?

Q.4 What states brings in the highest sales and profits ?

Q.5 Average profit by discount percentage.

Q.6 What category generates the highest sales and profits ?

Q.7 What sub-category generates the highest sales and profits ?

Q.8 What segment makes the most of our profits and sales ?

Q.9 What are the names of the products that are the most profitable to us?

Q.10 How many customers do we have in total?

LET'S START

TO SOLVE

THE QUESTIONS

Q.1 Which cities or regions are the most profitable ?

Query :

```
select city, round(sum(sales),2) as total_sales,
round(sum(profit),2) as total_profit
from superstore
group by city
order by total_profit desc;
```

Result Grid :

Result Grid | Filter Rows:

city	total_sales	total_profit
New York City	255248.97	61624.06
Los Angeles	173168.87	29806.92
Seattle	117772.58	28869
San Francisco	110917.04	17176.67
Detroit	42302.9	13117.05
Lafayette	25001.83	10001.55
Jackson	24903.45	7552.68
Atlanta	17197.84	6993.66
Minneapolis	16870.54	6824.58
San Diego	47115.06	6309.3
Springfield	42791.74	6078.02

Result 1

Q.2 What are the total profits and total sales per city ?

Query :

```sql
SELECT
 city,
SUM(sales) AS total_sales,
SUM(profit) AS total_profit
FROM superstore
GROUP BY  city
ORDER BY  city;
```

Result Grid :

Result Grid | Filter Rows: | Export:

city	total_sales	total_profit
Aberdeen	25.5	6.63
Abilene	1.392	-3.7584
Akron	2729.9860000000003	-186.6356
Albuquerque	2220.16	634.0880999999999
Alexandria	[illegible]0000000001	300.94710000000003
Allen	290.20599999999996	-39.8775
Allentown	853.252	-226.4504
Altoona	16.032	2.2044
Amarillo	3773.0628	-387.9683
Anaheim	7955.890000000001	1220.0626
Andover	435.84999999999997	124.1886
Ann Arbor	[illegible]	[illegible]

Result 5 ×

Q.3 What region generates the highest sales and profits ?

Query :

```
select region, sum(sales) as total_sales,
sum(profit) as total_profits
from superstore
group by region
order by total_profits desc;
```

Result Grid :

Result Grid | Filter Rows: | Export:

region	total_sales	total_profits
West	713471.3445000004	106021.14950000007
East	672194.0539999981	90672.01270000022
South	388983.58500000037	46035.68900000004
Central	497800.8728000007	40128.903

Insights :

We can observe that the west region is the one with the most sales and brings us in the highest profit.

The east region is pretty good looking for our company too.

Q.4 What states brings in the highest sales and profits ?

Query :

```sql
select state, sum(sales) as total_sales,
sum(profit) as total_profits,
round((sum(profit)/sum(sales))*100,2)
as profit_margin
from superstore
group by state
order by total_profits desc;
```

Result Grid :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

state	total_sales	total_profits	profit_margin
California	450567.5915000007	74669.20360000014	16.57
New York	309453.63299999974	73507.12520000007	23.75
Washington	136590.17199999996	32976.6151	24.14
Michigan	75991.58400000002	24340.948699999994	32.03
Virginia	70309.08999999998	18461.0153	26.26
Indiana	53474.43	18344.437500000004	34.31
Georgia	48296.30000000001	15933.646900000007	32.99
Kentucky	36591.74999999997	11199.696600000005	30.61
Minnesota	29863.149999999994	10823.1874	36.24
Delaware	27036.168999999994	9836.424200000001	36.38
New Jersey	34600.57199999999	9736.1716	28.14
[illegible]	[illegible]	[illegible]	[illegible]

Result 6

Insights :

The data shows the most profitable states. Besides we can see the total sales and profit margins.

Profit margins are important and it allows us to mostly think long-term as an investor to see potential big market.

Q.5 Average profit by discount percentage.

Query :

```
select discount,
round(avg(profit),2) as average_profit,
count(*) as order_count
from superstore
group by discount
order by discount;
```

Result Grid :

Result Grid | Filter Rows:

discount	average_profit	order_count
0	68.11	4657
0.1	96.06	94
0.15	27.29	52
0.2	25.2	3547
0.3	-45.68	227
0.32	-88.56	27
0.4	-112.51	205
0.45	-226.65	11
0.5	-310.7	66
0.6	-42.03	132
0.7	-101.39	391

Result 8

Insights :

Discounts may boost sales but reduce profits.

Identify discount ranges that maximize both sales and profit.

Q.6 What category generates the highest sales and profits ?

Query :

```sql
select category , sum(sales) as total_sales,
sum(profit) as total_profit,
round(sum(profit)/sum(sales)*100,2) as profit_margin
from superstore
group by category
order by total_profit desc;
```

Result Grid :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_sales	total_profit	profit_margin
Technology	835900.0669999964	145387.0965999999	17.39
Office Supplies	703502.9280000031	120489.8864000001	17.13
Furniture	733046.8612999996	16980.7711999998	2.32

Insights :

Out of the 3, it is clear that Technology and Office Supplies are the best in terms of profits.

Plus they seem like a good investment because of their profit margins.

Furnitures are still making profits but do not convert well in overall.

Q.7 What sub-category generates the highest sales and profits ?

Query :

```sql
select `sub-category`,sum(sales) as total_sales,
sum(profit) as total_profit,
round(sum(profit)/sum(sales)*100,2) as profit_margin
from superstore
group by `sub-category`
order by total_profit desc;
```

Result Grid :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sub-category	total_sales	total_profit	profit_margin
Copiers	149528.02999999994	55617.82490000001	37.2
Phones	329753.0880000001	44447.879100000006	13.48
Accessories	167380.3180000001	41936.63569999993	25.05
Paper	75356.11799999999	32712.168999999965	43.41
Binders	199905.71700000006	29983.021299999986	15
Chairs	328449.10300000076	26590.166300000026	8.1
Storage	216803.21200000012	21527.908299999996	9.93
Appliances	107532.161	18138.005399999995	16.87
Furnishings	82752.23000000004	11588.641999999987	14
Art	27118.791999999954	6527.786999999998	24.07
Envelopes	15339.489999999993	6460.8691000000035	42.12

Result 4

Insights :

Out of our 17 subcategories nationwide, our biggest profits comes from Copiers, Phones, Accessories and Paper.

The profits and profit margins on Copiers and Papers especially are interesting for the long run.

Our losses came from Tables, Bookcases and Supplies.

Q.8 What segment makes the most of our profits and sales ?

Query :

```
select segment,sum(sales) as total_sales,
sum(profit) as total_profit
from superstore
group by segment
order by total_profit desc;
```

Result Grid :

segment	total_sales	total_profit
Consumer	1150166.1819999903	132669.77519999977
Corporate	696604.5138000002	90366.3009000002
Home Office	425679.1605000003	59821.678100000085

Insights :

The consumer segment brings in the most profit followed by Corporate and then Home office.

Q.9 What are the names of the products that are the most profitable to us?

Query :

```
select `product name` , sum(sales) as total_sales,
sum(profit) as total_profit
from superstore
group by `product name`
order by total_profit desc;
```

Result Grid :

product name	total_sales	total_profit
Canon imageCLASS 2200 Advanced Copier	61599.824	25199.928000000004
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384	7753.039
Hewlett Packard LaserJet 3310 Copier	18839.686	6983.8836
Canon PC1060 Personal Laser Copier	11619.833999999999	4570.9347
HP Designjet T520 Inkjet Large Format Printer -...	18374.895	4094.9766
Ativa V4110MDD Micro-Cut Shredder	7699.89	3772.9461
3D Systems Cube Printer, 2nd Generation, Mag...	14299.89	3717.9714000000004
Plantronics Savi W720 Multi-Device Wireless He...	9367.289999999999	3696.2819999999997
Ibico EPK-21 Electric Binding System	15875.916000000001	3345.2823
Zebra ZM400 Thermal Label Printer	6965.700000000001	3343.536
Honeywell Enviracaire Portable HEPA Air Cleane...	11304.439999999999	3247.0199999999995

Insights :

The Cubify CubeX 3D Printer Double Head Print, Lexmark MX611dhe Monochrome Laser Printer and the Cubify CubeX 3D Printer Triple Head Print are the products that operate the most at a loss.

We should take this into account if we are thinking about modifying our stock.

Q.10 How many customers do we have in total?

Query :

```sql
select count(DISTINCT `customer id`) as total_customers
from superstore ;
```

Result Grid :

total_customers
793

CONCLUSION

This project effectively transforms raw retail data from the Superstore dataset into actionable business insights using MySQL. By analyzing product performance, customer behavior, regional trends, discount impact, and shipping efficiency, it helps identify key areas for growth and optimization. The insights gained support data-driven decisions that enhance profitability, customer targeting, and operational strategy in a retail environment.

THANK

YOU